

Each question is worth 2 points except where noted. Keep on truckin'...

Circle True or False. Read each statement completely before answering.

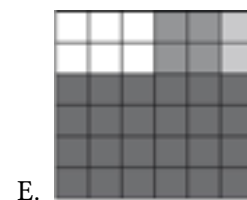
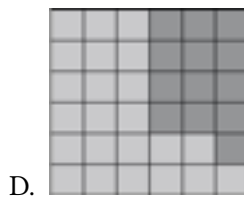
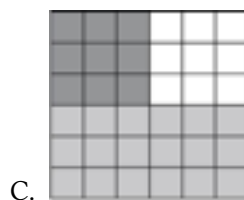
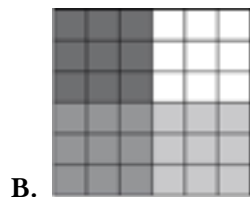
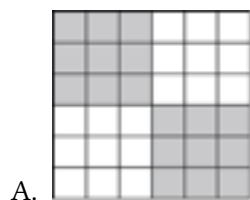
1. **TRUE** False The carrying capacity of a habitat may change as environmental conditions change.
2. **TRUE** False The brightly colored pattern of poison dart frogs is an example of aposomatic coloration.
3. **TRUE** False The intrinsic rate of population growth does not include immigration or emigration.
4. **TRUE** False A population may grow smaller if death rate is greater than immigration rate.
5. **TRUE** False Mutualism can be thought of as two species that ecologically exploit each other over the long term.
6. True **FALSE** Density-independent forces are most likely to affect population size when N is less than K .
7. True **FALSE** Evenness measures the distribution of relative abundance across all species in a population.
8. True **FALSE** Most organisms grow under idealized conditions because they are adapted to their environment.
9. **TRUE** False Population is the lowest hierarchical level for the biosphere category.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

10. Which of the following is *not* directly concerned with community ecology?

- A. **intraspecific competition**
- B. predation
- C. symbiosis
- D. resource partitioning
- E. herbivory

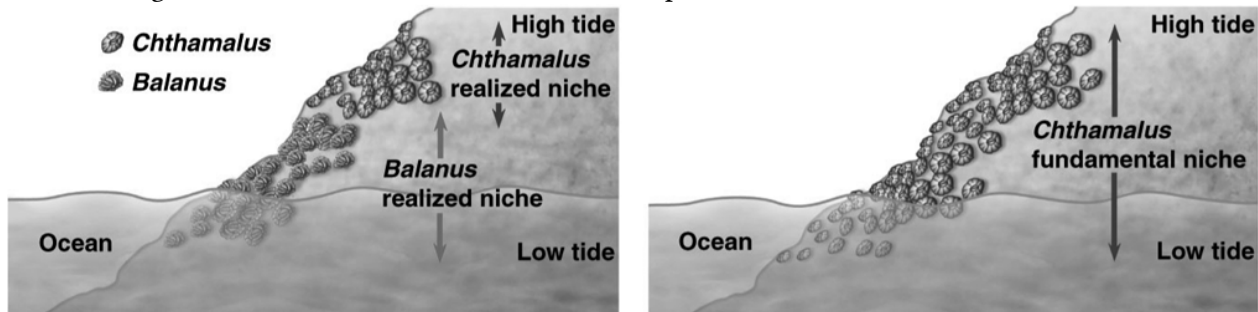
11. Which block in the figure below is most consistent intermediate levels of disturbance in an ecosystem? (Different shades, including white, represent different species. The small squares indicate relative abundance of individuals present.)



12. Given the following values and assuming exponential population growth, what will be the total population size after one generation has elapsed? Round to nearest whole individual.

$$\begin{array}{ll} b = 0.09 & N = 18,657 \\ d = 0.12 & K = 15,000 \end{array}$$

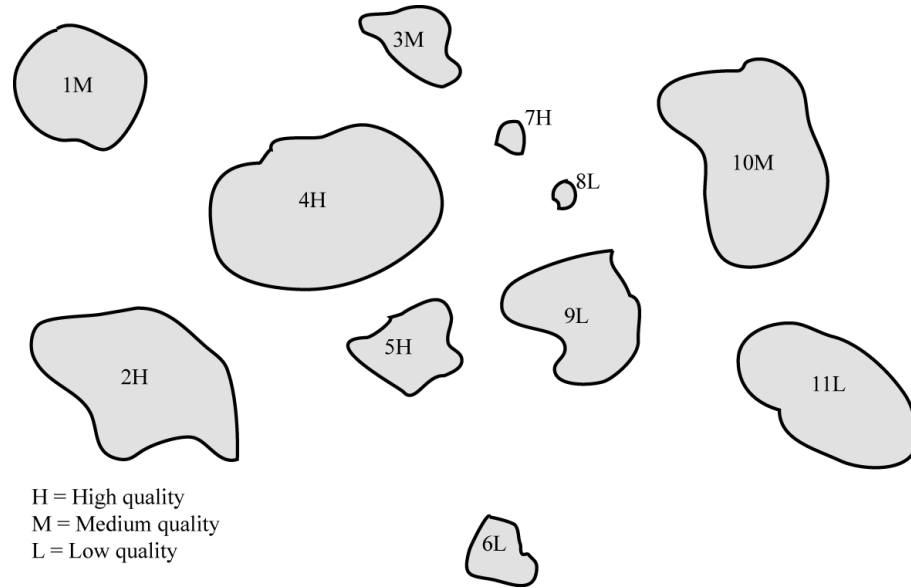
- A. 15,000
 B. 3,657
 C. 18,097
 D. 560
 E. 19,217
13. The dispersion pattern for individuals in across the entire metapopulation shown in question 33 is most likely
- A. uniform.
 B. even.
 C. **clumped.**
 D. random.
 E. gene flow.
14. In the experiment shown in the figure below, *Balanus* was removed from the habitat shown on the left. Which of the following statements is a valid conclusion of this experiment?



- A. *Balanus* is inferior to *Chthamalus* in competing for space on rocks lower in the intertidal zone.
 B. **The removal of *Balanus* shows that the realized niche of *Chthamalus* is smaller than its fundamental niche.**
 C. These two species of barnacle do not show competitive exclusion.
 D. If *Chthamalus* were removed, *Balanus*'s fundamental niche would become larger.
 E. *Balanus* can survive only in the lower intertidal zone because it is unable to resist dehydration.
15. Competitive exclusion means that
- A. one individual will always out compete another individual in a population.
 B. two species will partition resources to eliminate competition.
 C. a species will have a smaller realized niche than a fundamental niche.
 D. **one species will always out complete another species that occupies the same niche.**
 E. one population will always exclude another another population of the same species in a community.

16. Several species of warblers (a group of small colorful birds) in North America can live in the same tree only because they
- A. out-competed other species of birds, such as sparrows.
 - B. created different types of nests within the tree.
 - C. partitioned resources to eliminate competition.**
 - D. can fly to different heights within the tree.
 - E. eat different foods within the tree.
17. Cowbirds lay their eggs in the nests of other birds like cardinals. The cardinals raise the cowbird chicks as their own but the cowbird chicks usually kill the cardinal chicks of the other bird. This is most likely an example of
- A. resource partitioning.
 - B. symbiosis.
 - C. mutualism.
 - D. competition.
 - E. parasitism.**
18. Which of the following is most likely to contribute to density-dependent regulation of populations?
- A. Fires.
 - B. Floods.
 - C. Intraspecific competition for nutrients.**
 - D. The removal of toxic waste by decomposers.
 - E. Earthquakes.
19. Color that helps hide an animal in its environment is called
- A. hideandseek coloration.
 - B. environmental coloration.
 - C. mimicry.
 - D. aposomatic coloration
 - E. warning coloration.
 - F. cryptic coloration.**
20. The “role” or “job” of a species in a community, represented by all of the resources needed by the species, is called
- A. the niche.**
 - B. a predator.
 - C. a producer or consumer.
 - D. the resource.
 - E. a predator or prey.
-

33. (15 points) Consider the following metapopulation. Identify *one* subpopulation that is *most likely* to have the highest extinction rate and lowest recolonization rate. Justify your choice by explaining how patch size, distance and habitat quality interact to affect extinction and recolonization in your chosen patch. There may be more than one correct choice but explain for *only* one subpopulation.



Here are four numbers: 70, 47, 54, 38.

34. (9 points) Calculate the mean and standard deviation for the numbers listed above. **Show your work for all steps** or you will not get any pudding. Round your final answers to two digits after the decimal. 2 points per blank plus 5 points for your work.

$$s = \sqrt{\frac{\sum (Y - \bar{Y})^2}{n-1}}$$

$$\bar{Y} = \mathbf{52.25}$$

$$s = \mathbf{13.52}$$

35. (6 points) Calculate the Shannon diversity index for the same four numbers above. You may use the $\ln$ or $\log$ key on your calculator. Show your work and round to two digits after the decimal for your final answer. 2 points per blank plus 4 points for your work.

$$H' = - \sum p_i \ln p_i$$

$$H' = \mathbf{1.36 \text{ or } 0.59}$$

Give them 2 points for each correct answer, even if they do not show their work.

36. (6 points) You performed a t -test to compare a variable between two populations. The sample size from *each* population was 4 individuals. You calculated $t = 2.39$. Using $\alpha = 0.05$, do you accept or reject the null hypothesis? Explain how you used the table at right to decide, including specific numbers.

Table 1. Critical values of t .

df	α			
	0.10	0.05	0.02	0.01
1	6.31	12.71	31.82	63.66
2	2.92	4.30	6.96	9.92
3	2.35	3.18	4.54	5.84
4	2.13	2.78	3.75	4.60
5	2.02	2.57	3.36	4.03
6	1.94	2.45	3.14	3.71
7	1.89	2.36	3.00	3.50
8	1.86	2.31	2.90	3.36
9	1.83	2.26	2.82	3.25
10	1.81	2.23	2.76	3.17
11	1.80	2.20	2.72	3.11
12	1.78	2.18	2.68	3.06

Accept null hypothesis. Degrees of freedom is 10, so critical $t = 2.23$. Calculated $t = 2.21$, which is lower than the table value. If the calculated value is lower than the tabled value, then you accept (or fail to reject) the null hypothesis.

